Symbols used in 'Deflating degrees of freedom: dirac16complex when x_5, x_6, x_7 deflate' and in deflation.nb

1. Coordinates, metric and deflation

symbol	meaning	definition
$x_0, \ldots, x_7$	coordinates	x_0 hidden space; x_1, x_2, x_3 3-space; x_4 time; x_5, x_6, x_7 superluminal time, here the deflating degrees of freedom
i; j	index ranges	$i = 1, 2, 3$; $j = 5, 6, 7$
A, B, C	frame indices	$0, \ldots, 7$
$\mu, \nu, \kappa, \lambda, \rho$	coordinate indices	$0, \ldots, 7$; a λ carrying no index position is the quartic coupling of Section 4
η_{AB}, η^{AB}	flat metric	$\mathrm{diag}(+1, +1, +1, +1, -1, -1, -1, -1)$
H	Hubble parameter of the primordial field	$H > 0$
z	scaled hidden coordinate	$z = 6Hx_0$, $0 < z < \pi/2$, i.e. $0 < x_0 < \pi/(12H)$
$a_4(Hx_4)$	function in the vielbein of Pregravity6plane.nb	$a_4(Hx_4) = Hx_4$ for the primordial field
a	deflation constant	$a > 0$
b_{55}, b_{66}, b_{77}	deflation amplitudes	$b_{jj} > 0$
g_{55}, g_{66}, g_{77}	deflating metric functions (positive)	$g_{jj} = b_{jj} e^{-ax_4}$
${e^A}_\mu$, e_A	vielbein	$\mathrm{diag}(e_0, \ldots, e_7)$: $e_0 = \tan z$, $e_{1,2,3} = e^{-Hx_4} \sin^{-1/6} z$, $e_4 = 1$, $e_j = \sqrt{b_{jj}}\, e^{-ax_4/2}$
${E_A}^\mu$	inverse vielbein	${E_A}^\mu = \delta^\mu_A / e_A$
$g_{\mu\nu}$, $g^{\mu\nu}$	metric tensor and its inverse	$g_{\mu\nu} = \eta_{AB} {e^A}_\mu {e^B}_\nu = \mathrm{diag}(\tan^2 z, e^{-2Hx_4} \sin^{-1/3} z$ (three times), $-1, -g_{55}, -g_{66}, -g_{77})$
ds^2	line element	$g_{\mu\nu} dx^\mu dx^\nu$
$\sqrt{\lvert g \rvert}$	volume element	$e_0 \cdots e_7 = \sqrt{b_{55} b_{66} b_{77}} \tan z \sin^{-1/2} z \times e^{-3(H + a/2)x_4}$; $(\sqrt{\lvert g \rvert})^2 = \lvert \det g_{\mu\nu} \rvert$
L_j	proper size of Δx_j	$L_j = \sqrt{g_{jj}}\, \Delta x_j$; $d \ln L_j / dx_4 = -a/2$
pure deflation field	deflation alone	$ds^2 = dx_0^2 + dx_1^2 + dx_2^2 + dx_3^2 - dx_4^2 - g_{55} dx_5^2 - g_{66} dx_6^2 - g_{77} dx_7^2$

2. Dirac matrices, the field and its sectors

symbol	meaning	definition
I_n	identity matrix	$n \times n$
δ_{pq}, δ^μ_A	Kronecker symbol	1 if the indices are equal, else 0
ϵ_{hpq4}	Levi-Civita symbol	totally antisymmetric, $\epsilon_{1234} = 1$
σ	8x8 spinor metric	$\begin{pmatrix} 0 & I_4 \\ I_4 & 0 \end{pmatrix}$
s_h, t_h $(h = 1, 2, 3)$	self-dual, anti-self-dual 4x4 matrices	$(s_h)_{pq} = \epsilon_{hpq4} - \delta_{p4}\delta_{qh} + \delta_{ph}\delta_{q4}$, $(t_h)_{pq} = \epsilon_{hpq4} + \delta_{p4}\delta_{qh} - \delta_{ph}\delta_{q4}$
τ_A	real 8x8 matrices	$\tau_0 = I_8$; $\tau_h = \begin{pmatrix} 0 & s_h \\ s_h & 0 \end{pmatrix}$; $\tau_4, \tau_5, \tau_6 = \begin{pmatrix} 0 & t_k \\ -t_k & 0 \end{pmatrix}$ with $k = 3, 2, 1$; $\tau_7 = \tau_1 \cdots \tau_6$
$\bar\tau_A$	barred matrices	$\bar\tau_A = \sigma (\sigma \tau_A)^T = \sigma \tau_A^T \sigma$
$T^A = (T16^A)[A]$	real 16x16 Dirac matrices	$\begin{pmatrix} 0 & \bar\tau_A \\ \tau_A & 0 \end{pmatrix}$, $T^A T^B + T^B T^A = 2\eta^{AB} I_{16}$
T_A	lowered Dirac matrices	$T_A = \eta_{AB} T^B$

Γ (no indices)	chirality matrix	$T^0T^1\cdots T^7 = \mathrm{diag}(-I_8, I_8)$
P_L, P_R	chiral projectors	$\frac{1}{2}(I_{16}-\Gamma)$, $\frac{1}{2}(I_{16}+\Gamma)$
σ_{16}	Dirac-adjoint matrix	$T^0T^1T^2T^3 = \mathrm{diag}(-\sigma,\sigma)$, symmetric, $(\sigma_{16}T^A)^T = -\sigma_{16}T^A$
S^{AB}	Spin(4,4) generators	$\frac{1}{4}[T^A,T^B]$, $[S^{AB},T^C] = \eta^{BC}T^A - \eta^{AC}T^B$
ψ = dirac16complex	16-component complex Grassmann spinor field (second quantized)	$\psi = (\psi_0,\ldots,\psi_{15})^T$, $\psi_a\psi_b = -\psi_b\psi_a$, $\psi_a\psi_b^* = -\psi_b^*\psi_a$
θ, ζ	arbitrary Grassmann elements	$(\theta\zeta)^* = \zeta^*\theta^*$
ψ^*, $\psi^\dagger$	complex conjugate, conjugate transpose	componentwise conjugation; $\psi^\dagger = (\psi^*)^T$
$\bar\psi$	Dirac adjoint	$\psi^\dagger\sigma_{16}$
universe	mode function	a c-number solution of the linear field equation
K	Krein operator	$-i\sigma_{16}T^4$, $K^\dagger = K$, $K^2 = I_{16}$, $\sigma_{16}T^4 = iK$
β	Hermitian involution	$-iT^4$; $\sigma_{16} = K\beta$
$\Pi_\pm$	sector projectors	$\frac{1}{2}(I_{16}\pm K)$
$\psi_\pm$	mirror sectors	$\Pi_\pm\psi$; $K=+1$: +mass sector, $K=-1$: -mass sector
ω	frequency	mode $\propto e^{-i\omega x_4}$; energy density of a mode $\omega\psi^\dagger K\psi$
u	constant spinor	amplitude of a mode

3. Curvature and geodesics

symbol	meaning	definition
$\Gamma^\lambda{}_{\mu\nu}$	Christoffel symbols	$\frac{1}{2}g^{\lambda\rho}(\partial_\mu g_{\rho\nu}+\partial_\nu g_{\rho\mu}-\partial_\rho g_{\mu\nu})$
$R^\kappa{}_{\lambda\mu\nu}$	Riemann tensor	$\partial_\mu\Gamma^\kappa{}_{\lambda\nu} - \partial_\nu\Gamma^\kappa{}_{\lambda\mu} + \Gamma^\kappa{}_{\mu\rho}\Gamma^\rho{}_{\lambda\nu} - \Gamma^\kappa{}_{\nu\rho}\Gamma^\rho{}_{\lambda\mu}$
$R_{\kappa\lambda\mu\nu}$	lowered Riemann tensor	$g_{\kappa\rho}R^\rho{}_{\lambda\mu\nu}$
$R_{\lambda\nu}$	Ricci tensor	$R^\kappa{}_{\lambda\kappa\nu}$
R	Ricci scalar	$g^{\lambda\nu}R_{\lambda\nu}$
$G_{\mu\nu}$, $G^\mu{}_\nu$	Einstein tensor	$R_{\mu\nu}-\frac{1}{2}Rg_{\mu\nu}$; $G^\mu{}_\nu = g^{\mu\rho}G_{\rho\nu}$
$R_{\kappa\lambda\mu\nu}R^{\kappa\lambda\mu\nu}$	Kretschmann scalar	diagonal metric: $\sum\frac{g_{\kappa\kappa}}{g_{\lambda\lambda}g_{\mu\mu}g_{\nu\nu}}(R^\kappa{}_{\lambda\mu\nu})^2$
$\mathbb{R}^4$	flat 4-space	spanned by $x_0,\ldots,x_3$ in the pure deflation field
H^4	hyperbolic 4-space of radius $2/a$	metric $\frac{4}{a^2}\frac{dw^2+dy_5^2+dy_6^2+dy_7^2}{w^2}$, $w>0$
w, y_j	Poincare coordinates of the block $(x_4,\ldots,x_7)$	$w = \frac{2}{a}e^{ax_4/2}$, $y_j = \sqrt{b_{jj}}\,x_j$
ℓ	affine parameter of a world line	$\dot x = dx/d\ell$
p_5	conserved momentum	$p_5 = g_{5\mu}\dot x^\mu = -g_{55}\dot x_5$
x_4^*	turning point	world line: $\frac{1}{a}\ln\frac{b_{55}}{p_5^2}$; universe: $\frac{1}{a}\ln\frac{m^2+\cot^2 z\,k_0^2}{\bar q^{\,2}}$; numerical example: $\ln 8$
w_*, y_{5c}	radius and centre of a semicircle	$w_* = \frac{2\sqrt{b_{55}}}{a\lvert p_5\rvert}$; $(y_5-y_{5c})^2+w^2 = w_*^2$

4. Field equation, universes and pairs

symbol	meaning	definition
γ^μ	curved-space Dirac matrices	$E_A{}^\mu T^A$
$\omega_\mu{}^A{}_B$	spin-connection coefficients	$e^A{}_\nu(\partial_\mu E_B{}^\nu + \Gamma^\nu{}_{\mu\lambda}E_B{}^\lambda)$
Ω_μ	spinor connection	$\frac{1}{4}\omega_\mu{}^A{}_B T_A T^B$; here $\gamma^\mu\Omega_\mu = -\frac{3}{2}H\cot^2 z\,T^0 - \frac{3}{4}(a+2H)T^4$
D_μ	covariant derivative	$D_\mu\psi = \partial_\mu\psi + \Omega_\mu\psi$, $D_\mu\bar\psi = \partial_\mu\bar\psi - \bar\psi\Omega_\mu$
$\mathcal{L}$, $\mathcal{L}_m$	Lagrangian density (with mass m)	$\sqrt{\lvert g\rvert}[\frac{1}{2}(\bar\psi\gamma^\mu D_\mu\psi - (D_\mu\bar\psi)\gamma^\mu\psi) - m\bar\psi\psi - \frac{\lambda}{2}(\bar\psi\psi)^2]$

m	mass	real constant
λ	quartic self-coupling	real constant
$\partial^L/\partial\psi_a^*$	left Grassmann derivative	removes ψ_a^* from the left; sign $(-1)^{\text{position}-1}$
$\mathcal{C}$	mass-sign map	$\mathcal{C}\psi = \Gamma\psi^*$; $\mathcal{L}_m[\Gamma\psi^*] = \mathcal{L}_{-m}[\psi]$
P_1, R_1	mass-sign map, reflection	$P_1\psi(x) = T^1\psi(R_1 x)$, $R_1 : x_1 \mapsto -x_1$
ρ_s	mean-field scalar density	$\langle\bar\psi\psi\rangle$; effective mass $m + \lambda\rho_s$
F	friction-removing factor	$\sin^{1/4} z\, e^{3(a+2H)x_4/4}$, $\gamma^\mu\partial_\mu F + F\gamma^\mu\Omega_\mu = 0$
χ, χ'	rescaled spinor, its x_4-derivative	$\psi = F\chi$ (full field), $\psi = e^{3ax_4/4}e^{i(k_1x_1+q_5x_5)}\chi$ (pure field); $\chi' = d\chi/dx_4$
$k_0,\dots,k_3$; q_5, q_6, q_7; k_A	wave numbers	mode $e^{i(k_0x_0+\cdots+q_7x_7-\omega x_4)}u$; $k_5, k_6, k_7 \equiv q_5, q_6, q_7$, $k_4 = 0$
c_A	frozen coefficients	$c_A = E_A{}^A = 1/e_A$
S, s^2	spatial Dirac operator, its square	$S = \sum_{A\neq4} c_A k_A T^A$, $S^2 = s^2 I_{16}$, $s^2 = \sum_{A\neq4}\eta_{AA}(c_A k_A)^2$
X	mode matrix	$iS - i\omega T^4$
$\bar q^{\,2}$	deflating momentum squared	$\frac{q_5^2}{b_{55}} + \frac{q_6^2}{b_{66}} + \frac{q_7^2}{b_{77}}$
a_c	critical deflation rate	$a_c = 2H$
C_{2H}	late-time coefficient at $a = 2H$	$(k_1^2+k_2^2+k_3^2)\sin^{1/3} z - \bar q^{\,2}$
$c(x_4)$	coupling of the deflating direction x_5	$e^{ax_4/2}/\sqrt{b_{55}}$
κ_A	real coefficients	$\kappa_A = c_A k_A$ (frozen modes); $\kappa_1 = k_1$, $\kappa_5 = q_5 c(x_4)$ (pure field)
M	evolution matrix	$T^4(i\sum_{A\neq4}\kappa_A T^A - m)$; pure field: $T^4(ik_1T^1 + iq_5c(x_4)T^5 - m)$; $\chi' = M\chi$
h_0	Hamiltonian at $c = 0$	$-k_1T^4T^1 - imT^4$, $\chi' = -ih_0\chi$ at $c = 0$
N	norm	$\chi^\dagger\chi = P_+ + P_-$
$P_\pm$	weights of the +mass and -mass sectors	$\chi^\dagger\Pi_\pm\chi$
Q	Krein charge	$\chi^\dagger K\chi = P_+ - P_-$, conserved: $M^\dagger K + KM = 0$
$\bar\psi\gamma^4\psi$	x_4-component of the U(1) current	$\psi^\dagger\sigma_{16}T^4\psi = i\psi^\dagger K\psi$

5. Names used in deflation.nb

notebook name	symbol	definition
`X`, `eta4488`	$(x_0,\dots,x_7)$, η_{AB}	coordinates; $\mathrm{diag}(1,1,1,1,-1,-1,-1,-1)$
`ID4`, `ID8`, `ID16`	I_4, I_8, I_{16}	identity matrices
`sigma`, `s4by4[h]`, `t4by4[h]`	σ, s_h, t_h	Section 2 above
`Qa`, `Qb`	parts of s_h, t_h	ϵ_{hpq4}; $\delta_{p4}\delta_{qh} - \delta_{ph}\delta_{q4}$; $s_h = Qa - Qb$, $t_h = Qa + Qb$
`sixAntiSymmetric8by8`	$\tau_1,\dots,\tau_6$	Section 2 above
`tau[A]`, `taubar[A]`	τ_A, $\bar\tau_A$	Section 2 above
`T16A[A]` $(A = 0..7)$	T^A	the matrices (T16^A)[A] of Pregravity6plane.nb
`T16A[8]`, `Gamma5`	Γ	$T^0\cdots T^7$
`sigma16`, `PL`, `PR`, `SAB`	σ_{16}, P_L, P_R, S^{AB}	Section 2 above
`gen["psi", a, d]`, `gen["psiS", a, d]`	$\partial_d\psi_a$, $\partial_d\psi_a^*$	Grassmann generators; `d` is the sorted list of derivative indices
`psi`, `psiS`, `dpsi[mu]`, `dpsiS[mu]`	ψ, ψ^*, $\partial_\mu\psi$, $\partial_\mu\psi^*$	vectors of generators
`gClean`, `gSortSign`, `gFromMono`	bookkeeping	drop zero coefficients; sort generators with the sign of the permutation (zero if a generator repeats); build a monomial
`gAdd`, `gScale`, `gMul`, `gGen`, `gNum`	algebra operations	sum, multiplication by a number, Grassmann product, a generator, a number
`gDL`, `gDR`	left, right derivative	remove a generator at position p of k with sign $(-1)^{p-1}$, resp. $(-1)^{k-p}$

`gDx[mu, x, X]`	total derivative d/dx_μ	Leibniz rule; a generator with derivative list d goes to $d \cup \{\mu\}$
`gConj`	complex conjugation	$(c\,\theta_1 \cdots \theta_k)^* = c^* \theta_k^* \cdots \theta_1^*$
`gBil[u, M, v]`, `gMatVec`, `gVecAdd`, `gVecScale`	bilinear form and vector operations	$\sum_{ab} u_a M_{ab} v_b$; Mv; $u+v$; cu
`gZeroQ`, `gSimplify`, `gLinSub`	tests and substitution	element is zero; simplify coefficients; substitute each generator by a linear combination
`Sbil`	$\bar\psi\psi$	`gBil[psiS, sigma16, psi]`
`K`, `betaKS`, `Pp`, `Pm`	K, β, Π_+, Π_-	Section 2 above
`uv`, `om`, `psiM`, `enM`	u, ω, $e^{-i\omega x_4}u$, mode energy density	$u =$ `ur` + i `ui` with real components
`spinConnection[e, X]`	Ω_μ, $\Gamma^\lambda{}_{\mu\nu}$, γ^μ, $g_{\mu\nu}$	for a diagonal vielbein `e`
`covConst`	covariant constancy test	$\partial_\mu\gamma^\nu + \Gamma^\nu{}_{\mu\lambda}\gamma^\lambda + [\Omega_\mu, \gamma^\nu] = 0$
`christoffelG`, `riemannG`, `ricciG`, `einsteinG`, `kretschmannG`	$\Gamma^\lambda{}_{\mu\nu}$, $R^\kappa{}_{\lambda\mu\nu}$, $R_{\lambda\nu}$, $G_{\mu\nu}$, $R_{\kappa\lambda\mu\nu}R^{\kappa\lambda\mu\nu}$	Section 3 above (`kretschmannG` general metric)
`kretschmannDiag`	Kretschmann scalar	diagonal-metric formula of Section 3 above
`z0`, `gjj`, `eDefl`, `gDefl`	z, (g_{55}, g_{66}, g_{77}), $(e_0, \ldots, e_7)$, $g_{\mu\nu}$	Section 1 above
`assumD`	assumptions	$H > 0$, $0 < x_0 < \pi/(12H)$, $a > 0$, $b_{jj} > 0$, real coordinates, real m and λ
`sqrtgD`	$\sqrt{\lvert g\rvert}$	$e_0 \cdots e_7$
`RicD`, `RsD`, `GmixD`, `KD`	$R_{\mu\nu}$, R, $G^\mu{}_\mu$, Kretschmann scalar	full deflation field
`gPure`, `RicP`, `RsP`	pure deflation metric, its $R_{\mu\nu}$ and R	$\mathrm{diag}(1,1,1,1,-1,-g_{55},-g_{66},-g_{77})$
`Y4`, `g4`, `RP`, `R4`, `R4low`	block coordinates and metric, Riemann tensors	(x_4, x_5, x_6, x_7), $\mathrm{diag}(-1,-g_{55},-g_{66},-g_{77})$; $R^\kappa{}_{\lambda\mu\nu}$ of the pure field and of the block; $R_{\kappa\lambda\mu\nu}$ of the block
`xOfY`, `JacY`, `gW`, `w`, `y5`, `y6`, `y7`	Poincare map	$x_4 = \frac{2}{a}\ln\frac{aw}{2}$, $x_j = y_j/\sqrt{b_{jj}}$; Jacobian; block metric in (w, y_5, y_6, y_7)
`p5`, `x5dot`, `x4dot2`, `Gam45`	p_5, $\dot x_5$, $\dot x_4^2$, Christoffel symbols of (x_4, x_5)	Section 3 above
`wOf`, `wstar`	$w(x_4)$, w_*	$\frac{2}{a}e^{ax_4/2}$, $2\sqrt{b_{55}}/(ap_5)$
`scD`, `OmD`, `gamD`, `spinD`	spin connection data, Ω_μ, γ^μ, $\gamma^\mu\Omega_\mu$	full deflation field
`LD`, `realRuleD`	$\mathcal{L}$, reality rules	`realRuleD` sets $m^* = m$, $\lambda^* = \lambda$, $H^* = H$, $a^* = a$, $b_{jj}^* = b_{jj}$, $x_0^* = x_0$, $x_4^* = x_4$
`ELD`, `DpsiD`, `expectedD`	Euler-Lagrange expressions, $\gamma^\mu D_\mu\psi$, expected form	expected form $\sqrt{\lvert g\rvert}\sigma_{16}(\gamma^\mu D_\mu\psi - m\psi - \lambda(\bar\psi\psi)\psi)$
`ruleC`, `ruleP`	substitutions for $\mathcal{C}$ and P_1	$\psi_a \mapsto \Gamma_{ab}\psi_b^*$, $\psi_a^* \mapsto \Gamma_{ab}\psi_b$; $\psi_a \mapsto T^1_{ab}\psi_b$ with $\partial_1 \mapsto -\partial_1$
`FD`, `cD`, `kv`, `Sslash`, `omega2`	F, c_A, k_A, S, ω^2	Section 4 above
`qq`, `om0`, `x4starD`, `om2H`, `C2H`	$\bar q^{\,2}$, ω^2 at $k_1 = k_2 = k_3 = 0$, x_4^*, ω^2 at $a = 2H$, C_{2H}	Section 4 above
`sampleD`	sample values	$H = 1$, $x_0 = 1/10$, $m = 1$, $k_0 = 0$, $k_1 = 1$, $k_2 = k_3 = 0$, $q_5 = 1$, $q_6 = q_7 = 0$, $b_{jj} = 1$
`ePure`, `scP`	pure-field vielbein and spin connection	$(1,1,1,1,1, \sqrt{b_{55}}e^{-ax_4/2}, \sqrt{b_{66}}e^{-ax_4/2}, \sqrt{b_{77}}e^{-ax_4/2})$
`chiS`, `phaseP`, `psiP`, `DpsiP`, `Mchi`	χ, phase, ψ, $\gamma^\mu D_\mu\psi - m\psi$, M	phase $e^{3ax_4/4}e^{i(k_1x_1+q_5x_5)}$
`aN`, `bN`, `kN`, `mN`, `qN`, `x4star`	a, b_{55}, k_1, m, q_5, x_4^* in the example	$1, 1, 1, 1, 1/2$; $\ln 8$
`h0`, `Kp`, `Km`, `vals0`, `vecs0`, `start`	h_0, Π_+, Π_-, eigen-data, $\chi(-8)$	eigenvector of $\Pi_+ h_0 \Pi_+ + 10\Pi_-$ with eigenvalue $\sqrt{2}$
`cf`, `chiF`, `eqs`, `solChi`, `chiAt`	numerical χ	components, equations $\chi' = M\chi$, `NDSolve` solution, $\chi(x_4)$
`normAt`, `minusFrac`, `plusW`, `minusW`	N, P_-/N, P_+, P_-	Section 4 above
`crossD[A]`	sector cross term of direction A	$(I_{16}+K)\sigma_{16}c_A T^A(I_{16}-K)$
`kapA`, `Mdef`, `MdefH`	κ_A, M, $M^\dagger$	$M = T^4(i\sum_{A\neq 4}\kappa_A T^A - m)$